

Splunk SOC Practice Lab

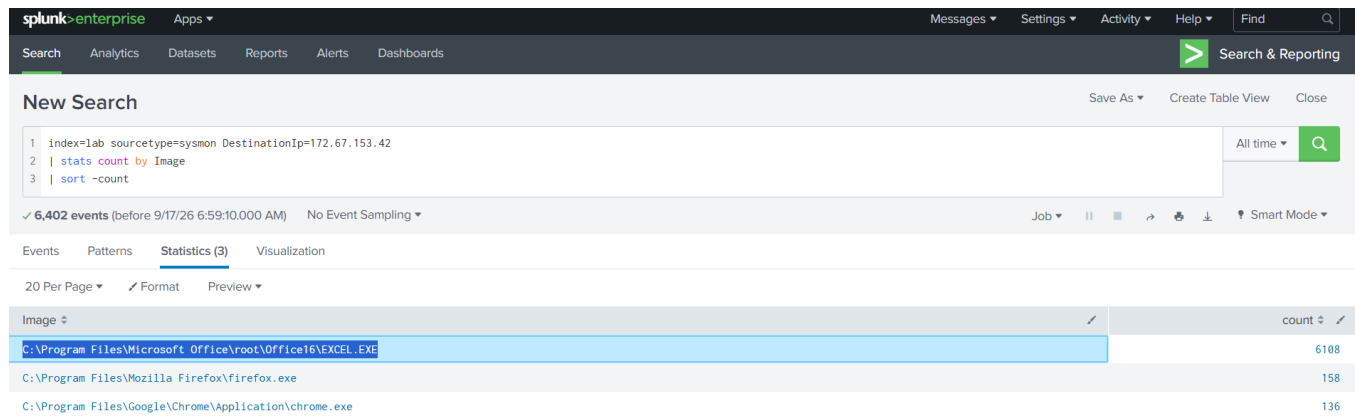
Practical documentation of Splunk searches, network traffic analysis, Sysmon correlation, enrichment and investigation workflow.

Area	Skills Practiced
Splunk SPL	index filtering, table, stats, sort, timechart, lookup, where
Network Analysis	FortiGate traffic, IPs, ports, protocols, bytes, duration
Endpoint Correlation	Sysmon process Image, ProcessId and ParentImage
Threat Context	ASN, organization, destination category and risk enrichment
SOC Investigation	Finding unusual volume, high-risk destinations and correlating logs

SOC reminder: A large traffic volume or a high-risk enrichment result is an investigation indicator, not automatic proof of malicious activity. Validate the destination, user, process and related logs before making a conclusion.

1. Network Traffic Query – Timechart by Destination

Used a timechart to compare traffic counts to selected destination IPs. This helps a SOC analyst understand communication patterns over time.



Evidence screenshot: Screenshot 2026-09-17 122957(2).png

2. FortiGate Traffic – Connection Details

Displayed timestamp, source/destination, port, protocol, bytes and duration from FortiGate logs. This is useful for examining individual network connections.

splunk>enterpriseApps

MessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

Search & Reporting

New Search

Save AsCreate Table ViewClose

1 index=lab sourcetype=zeek_conn service=quic

2 | lookup asn_lookup network AS "id.resp_h" OUTPUT dst_asn_org dst_category

3 | where duration > 3600

4 | table _time "id.orig_h" "id.resp_h" duration orig_bytes resp_bytes dst_asn_org

✓ 1 event (before 9/17/26 6:42:55.000 AM) No Event SamplingJob

EventsPatternsStatistics (1)Visualization

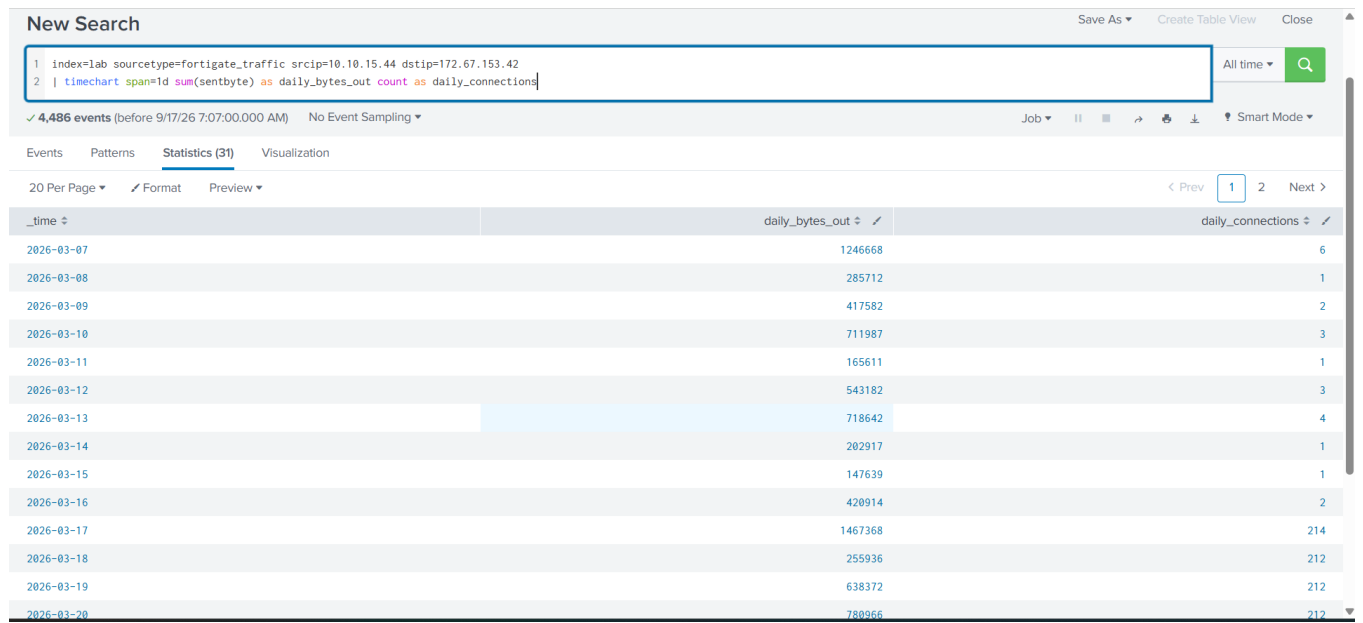
20 Per PageFormatPreview

_time	id.orig_h	id.resp_h	duration	orig_bytes	resp_bytes	dst_asn_org
2026-04-02 02:47:00	10.10.15.44	172.67.153.42	15180.0	847823104	2097152	Cloudflare Inc.

Evidence screenshot: Screenshot 2026-09-17 121418(2).png

3. Sysmon – Process and Network Correlation

Correlated FortiGate traffic with Sysmon process information to see which Windows process was associated with the connection.



Evidence screenshot: Screenshot 2026-09-17 123727(2).png

4. Sysmon – Process Frequency Analysis

Counted process images for traffic to a destination. A high count for one process can help identify the application generating repeated connections.

New Search

Save AsCreate Table ViewClose

1 index=lab sourcetype=fortigate_traffic

2 | lookup asn_lookup network AS dstip OUTPUT dst_asn_org dst_category dst_risk

3 | stats count by dst_category dst_asn_org

4 | sort -count

All time

✓ 18,187 events (before 9/17/26 6:35:26.000 AM) No Event SamplingJob || ↻ ⚙ ⬇ ⚡ Smart Mode

EventsPatternsStatistics (16)Visualization

20 Per PageFormatPreview

dst_category	dst_asn_org	count
CDN	Cloudflare Inc.	7501
Cloud Provider	Google LLC	2016
Cloud Provider	Microsoft Corporation	1545
CDN	Fastly Inc.	1268
Video Conferencing	Zoom Video Communications	794
CDN	Akamai Technologies	779
Cloud Provider	Amazon AWS	773
Cloud Provider	GitHub Inc.	753
Software Update	Canonical Group Limited	745
Unknown	Unknown	743
CDN	Amazon CloudFront	738
DNS Resolver	Cloudflare Inc. (DNS Resolver)	221

Evidence screenshot: Screenshot 2026-09-17 120634(2).png

5. FortiGate – Destination ASN and Category Analysis

Used ASN/category enrichment to understand the type and organization associated with destination networks. Enrichment adds context to raw IP addresses.

splunk>enterprise
Apps
Messages
Settings
Activity
Help
Find

Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Reporting

New Search

```
1 index=lab sourcetype=fortigate_traffic dstip=172.67.153.42
2 | lookup asn_lookup network AS dstip OUTPUT dst_asn dst_asn_org dst_category dst_risk
3 | stats count by dstip dst_asn dst_asn_org dst_category dst_risk
```

All time

4,877 events (before 9/17/26 7:06:25.000 AM)
No Event Sampling
Job
||
Smart Mode

Events
Patterns
Statistics (1)
Visualization

20 Per Page
Format
Preview

dstip	dst_asn	dst_asn_org	dst_category	dst_risk	count
172.67.153.42	AS13335	Cloudflare Inc.	CDN	low	4877

Evidence screenshot: Screenshot 2026-09-17 123649(2).png

6. FortiGate – Destination IP Enrichment

Grouped a destination IP with ASN, organization, category and risk information. This demonstrates how lookup data can add security context.

```

1 index=l4b sourcetype=fortigate-traffic
2 | lookup asn_lookup network AS dstip OUTPUT dst_asn_org dst_category dst_risk
3 | stats count by dst_category | dst_asn_org
4 | sort -count

```

✓ 18,187 events (before 9/17/26 7:02:08.000 AM) No Event Sampling ▼
Job ▼ || || → 📄 ⬇️ ⚙️ Smart Mode ▼

Events

Statistics (16)

Visualization

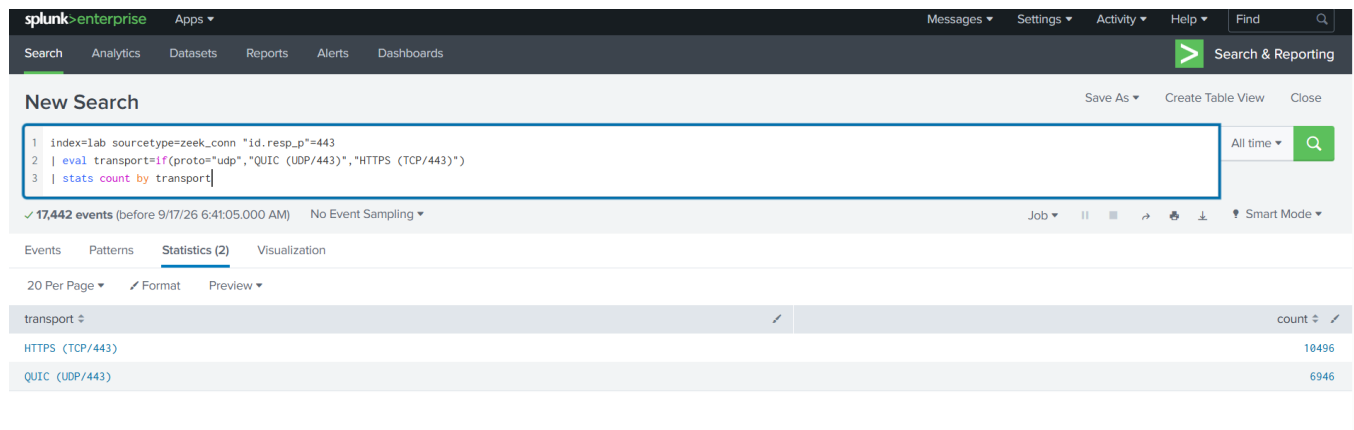
20 Per Page ▼
✍️ Format
Preview ▼

dst_category ⬇️ ✎	dst_asn_org ⬇️ ✎	count ⬇️ ✎
CDN	Cloudflare Inc.	7501
Cloud Provider	Google LLC	2016
Cloud Provider	Microsoft Corporation	1545
CDN	Fastly Inc.	1268
Video Conferencing	Zoom Video Communications	794
CDN	Akamai Technologies	779
Cloud Provider	Amazon AWS	773
Cloud Provider	GitHub Inc.	753
Software Update	Canonical Group Limited	745
Unknown	Unknown	743
CDN	Amazon CloudFront	738
DNS Resolver	Cloudflare Inc. (DNS Resolver)	221
DNS Resolver	Google LLC (DNS Resolver)	179
DNS Resolver	Quad9 (DNS Resolver)	122

Evidence screenshot: Screenshot 2026-09-17 123356(2).png

7. Network Traffic – Daily Bytes and Connections

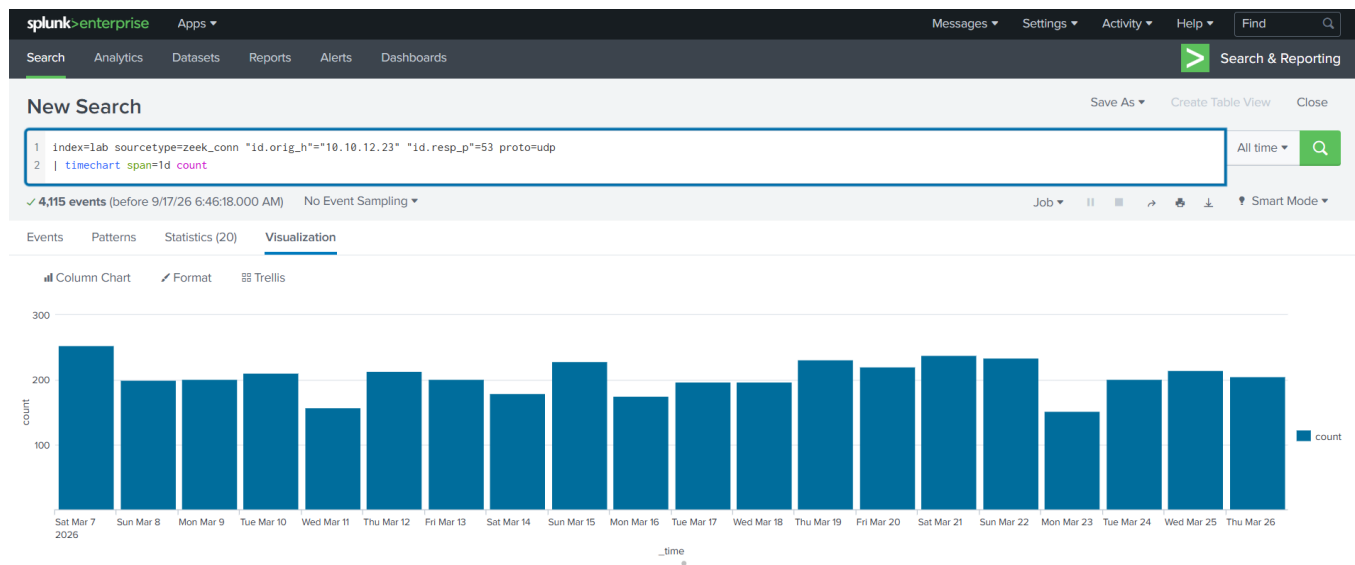
Calculated daily outbound bytes and daily connection counts. This can help identify changes or spikes in network activity.



Evidence screenshot: Screenshot 2026-09-17 121153(2).png

8. Network Traffic – Daily Visualization

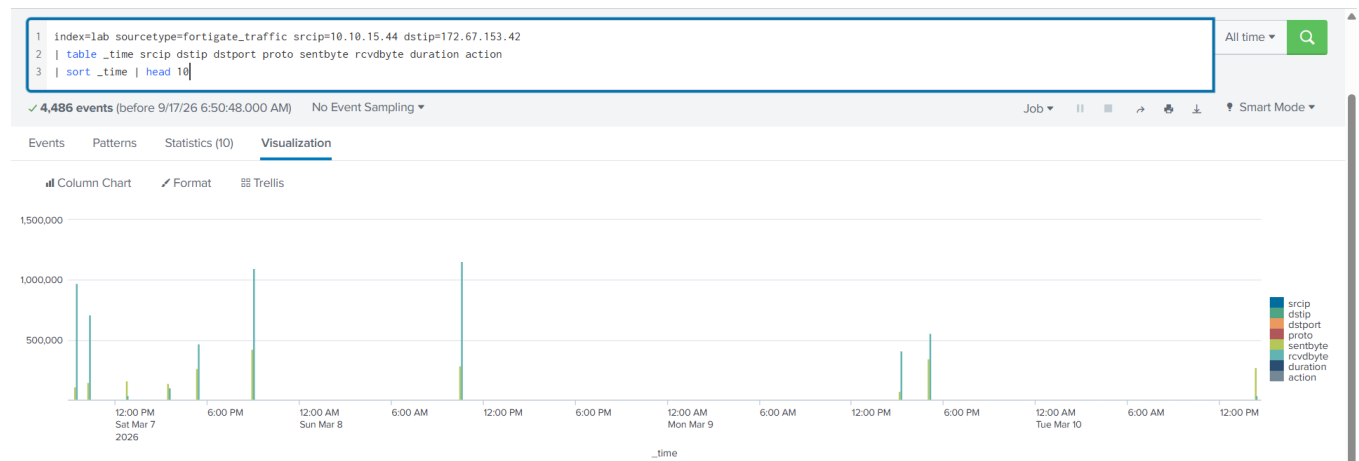
Visualized daily traffic volume and connection activity to make trends easier to spot during monitoring.



Evidence screenshot: Screenshot 2026-09-17 121637(2).png

9. FortiGate Traffic – High-Risk Destinations

Filtered for high/critical destination risk and displayed source, destination, port, protocol, byte counts and enrichment details. These results require investigation and validation.



Evidence screenshot: Screenshot 2026-09-17 122106(1).png

10. FortiGate Traffic – Destination IP Summary

Summarized a destination IP using ASN, organization, category and risk fields. The displayed destination was classified as a CDN with low risk in the lab data.

New Search

Save AsCreate Table ViewClose

1 index=lab sourcetype=fortigate_traffic

2 | lookup asn_lookup network AS dstip OUTPUT dst_asn dst_asn_org dst_category dst_risk

3 | where dst_risk="high" OR dst_risk="critical"

4 | table _time srcip srcname dstip dstport proto sentbyte rcvbyte dst_asn_org dst_category dst_risk

5 | sort _time

All time

✓ 10 events (before 9/17/26 7:04:06.000 AM) No Event SamplingJob||↻🔖📄Smart Mode

EventsPatternsStatistics (10)Visualization

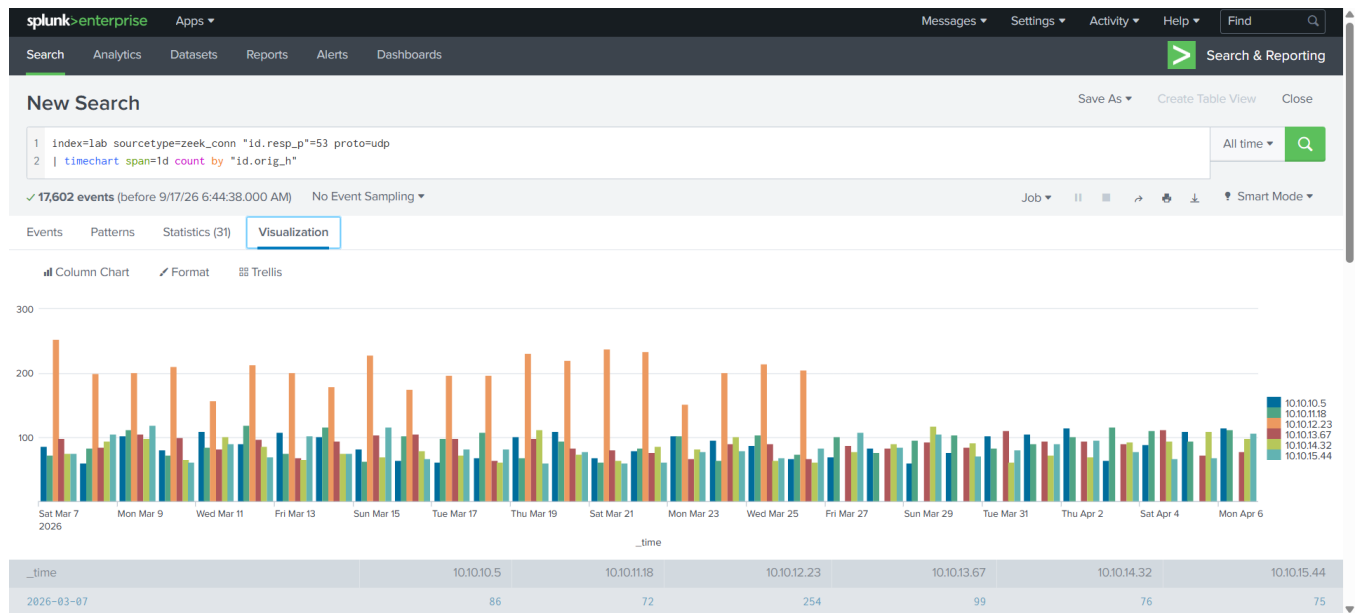
20 Per PageFormatPreview

_time	srcip	srcname	dstip	dstport	proto	sentbyte	rcvbyte	dst_asn_org	dst_category	dst_risk
2026-03-22 10:00:00	10.10.11.18	WKST-HR-02	185.220.101.34	443	6	2841	2650	F3 Netze e.V.	Tor Exit / Anonymisation	high
2026-03-22 10:02:00	10.10.11.18	WKST-HR-02	185.220.101.34	443	6	2956	2890	F3 Netze e.V.	Tor Exit / Anonymisation	high
2026-03-22 10:04:00	10.10.11.18	WKST-HR-02	185.220.101.34	443	6	3102	2700	F3 Netze e.V.	Tor Exit / Anonymisation	high
2026-03-22 10:06:00	10.10.11.18	WKST-HR-02	185.220.101.34	443	6	2778	2600	F3 Netze e.V.	Tor Exit / Anonymisation	high
2026-03-22 10:08:00	10.10.11.18	WKST-HR-02	185.220.101.34	443	6	3045	2980	F3 Netze e.V.	Tor Exit / Anonymisation	high
2026-03-22 10:10:00	10.10.11.18	WKST-HR-02	185.220.101.34	443	6	2901	2750	F3 Netze e.V.	Tor Exit / Anonymisation	high
2026-03-25 13:52:00	10.10.13.67	WKST-SALES-11	185.213.154.88	443	6	29763	74448	Stark Industries Solutions Ltd	Bulletproof Hosting	high
2026-03-25 14:06:00	10.10.13.67	WKST-SALES-11	185.213.154.88	443	6	29545	12777	Stark Industries Solutions Ltd	Bulletproof Hosting	high
2026-03-25 15:52:45	10.10.13.67	WKST-SALES-11	185.213.154.88	443	6	43446	59370	Stark Industries Solutions Ltd	Bulletproof Hosting	high
2026-03-25 17:34:59	10.10.13.67	WKST-SALES-11	185.213.154.88	443	6	28848	27401	Stark Industries Solutions Ltd	Bulletproof Hosting	high

Evidence screenshot: Screenshot 2026-09-17 123536(2).png

11. Network Traffic – Trend Visualization

Visualized network traffic trends over time. Trend analysis can help a SOC analyst notice unusual increases or changes in communication behavior.



Evidence screenshot: Screenshot 2026-09-17 121512(2).png

12. Network Traffic – High Outbound Data Transfer

Sorted FortiGate traffic by sent bytes. The top connection transferred about 847.8 MB outbound, making it a useful example of a high-volume connection to investigate.

New Search

Save AsCreate Table ViewClose

1 index=lab sourcetype=fortigate_traffic srcip=10.10.15.44 dstip=172.67.153.42 earliest="04/02/2026:00:00:00" latest="04/03/2026:00:00:00"

2 | table _time srcip dstip dstport proto sentbyte rcvdbyte duration

3 | sort -sentbyte

All time

✓ 212 events (before 4/3/26 12:00:00.000 AM) No Event SamplingJob Smart Mode

EventsPatternsStatistics (212)Visualization

20 Per PageFormatPreview

< Prev12345678...Next >

_time	srcip	dstip	dstport	proto	sentbyte	rcvdbyte	duration
2026-04-02 02:47:00	10.10.15.44	172.67.153.42	443	17	847823184	2897152	15180
2026-04-02 16:02:41	10.10.15.44	172.67.153.42	443	17	81492	276724	113
2026-04-02 09:19:46	10.10.15.44	172.67.153.42	443	17	1198	355	3
2026-04-02 18:47:57	10.10.15.44	172.67.153.42	443	17	1197	217	3
2026-04-02 13:08:17	10.10.15.44	172.67.153.42	443	17	1197	368	2
2026-04-02 20:24:19	10.10.15.44	172.67.153.42	443	17	1196	285	5
2026-04-02 15:04:09	10.10.15.44	172.67.153.42	443	17	1193	316	1
2026-04-02 21:19:54	10.10.15.44	172.67.153.42	443	17	1189	382	1
2026-04-02 11:47:32	10.10.15.44	172.67.153.42	443	17	1186	200	2
2026-04-02 15:27:40	10.10.15.44	172.67.153.42	443	17	1183	374	2
2026-04-02 16:32:11	10.10.15.44	172.67.153.42	443	17	1182	293	5
2026-04-02 21:36:02	10.10.15.44	172.67.153.42	443	17	1180	341	3
2026-04-02 13:43:41	10.10.15.44	172.67.153.42	443	17	1179	316	4

Evidence screenshot: Screenshot 2026-09-17 123934(1).png

13. Destination IP Enrichment – ASN and Risk

Used lookup enrichment to identify the ASN, organization, category and risk associated with a destination IP. This demonstrates IP reputation/context analysis.

spunk>enterprise
Apps ▾
Messages ▾
Settings ▾
Activity ▾
Help ▾
Find

Search
Analytics
Datasets
Reports
Alerts
Dashboards
➤ Search & Reporting

New Search

Save As ▾
Create Table View
Close

```

1 index=lab sourcetype=sysmon SourceIp=10.10.15.44 DestinationIp=172.67.153.42
2 | table _time SourceIp DestinationIp DestinationPort Image ProcessId ParentImage
3 | sort _time | head 10

```

All time ▾
🔍

✓ 6,140 events (before 9/17/26 6:54:37:000 AM)
No Event Sampling ▾
Job ▾
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Smart Mode ▾

Events
Patterns
Statistics (10)
Visualization

20 Per Page ▾
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Preview ▾

<u>_time</u> ▴ ▾	<u>SourceIp</u> ▴ ▾	<u>DestinationIp</u> ▴ ▾	<u>DestinationPort</u> ▴ ▾	<u>Image</u> ▴ ▾	<u>ProcessId</u> ▴ ▾	<u>ParentImage</u> ▴ ▾
2026-03-07 11:01:42	10.10.15.44	172.67.153.42	443	C:\Program Files\Mozilla Firefox\firefox.exe	9821	C:\\Windows\\explorer.exe
2026-03-07 11:01:42	10.10.15.44	172.67.153.42	443	C:\Program Files\Mozilla Firefox\firefox.exe	9821	
2026-03-08 16:25:18	10.10.15.44	172.67.153.42	443	C:\Program Files\Mozilla Firefox\firefox.exe	3097	C:\\Windows\\explorer.exe
2026-03-08 16:25:18	10.10.15.44	172.67.153.42	443	C:\Program Files\Mozilla Firefox\firefox.exe	3097	
2026-03-11 12:14:06	10.10.15.44	172.67.153.42	443	C:\Program Files\Google\Chrome\Application\chrome.exe	5098	
2026-03-11 12:14:06	10.10.15.44	172.67.153.42	443	C:\Program Files\Google\Chrome\Application\chrome.exe	5098	C:\\Windows\\explorer.exe
2026-03-12 17:55:42	10.10.15.44	172.67.153.42	443	C:\Program Files\Mozilla Firefox\firefox.exe	4976	
2026-03-12 17:55:42	10.10.15.44	172.67.153.42	443	C:\Program Files\Mozilla Firefox\firefox.exe	4976	C:\\Windows\\explorer.exe
2026-03-14 15:57:58	10.10.15.44	172.67.153.42	443	C:\Program Files\Mozilla Firefox\firefox.exe	9012	
2026-03-14 15:57:58	10.10.15.44	172.67.153.42	443	C:\Program Files\Mozilla Firefox\firefox.exe	9012	C:\\Windows\\explorer.exe

Evidence screenshot: Screenshot 2026-09-17 122501(2).png

Key Takeaways

- I practiced writing and modifying SPL searches rather than only viewing raw events.
- I learned how FortiGate traffic fields can be used to investigate network connections.
- I practiced using lookup enrichment to add ASN, organization, category and risk context.
- I learned how Sysmon process information can be correlated with network activity.
- I practiced using statistics and visualizations to identify traffic trends and high-volume connections.
- I learned to treat unusual activity as an investigation lead and validate it using multiple data sources.

Lab use: This document contains screenshots from an authorized training/lab environment and is intended for learning and portfolio documentation.